

Non-calculator Part:

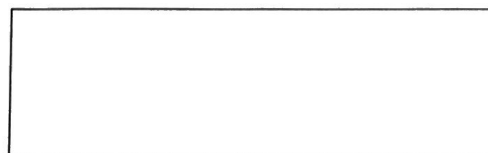
1. Find the inverse of matrix A given below.

$$A = \begin{bmatrix} 4 & 3 & 5 \\ -5 & -3 & -9 \\ -1 & 0 & -3 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$$

2. Use Gauss-Jordan elimination to solve the following system of equations. You should indicate what specific row operation(s) you perform in each step and show all your work clearly. Hint: Your Analysis teachers are being nice and the answers are integers!

$$\begin{aligned} 4x - 3y &= 15 \\ 2x + 5y &= 79 \end{aligned}$$



3. Consider a Markov Chain represented by the transition matrix T below. How would you find the equilibrium vector of this Markov Chain algebraically? Write a 3×4 matrix that you would enter into a calculator such that its reduced row echelon form yields the equilibrium vector.

Do **NOT** solve the system.

$$T = \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.1 & 0.4 & 0.5 \\ 0 & 0.6 & 0.4 \end{bmatrix}$$

$\left[\begin{array}{ccc c} \end{array} \right]$

Calculator Part:

Write down the calculation that you entered into your calculator to solve each problem.

4. A research group is studying career movement among adults in a certain region. At the end of each year, a person can be in one of three states: **T** = works in the tech industry; **N** = works in a non-tech industry; **U** = unemployed.

Historical data suggests the following transition probabilities from one year to the next:

- A person working in **tech** (T) will
 - stay in tech with probability 0.7
 - move to a non-tech job with probability 0.2
 - become unemployed with probability 0.1
- A person working in **non-tech** (N) will
 - switch to tech with probability 0.15
 - stay in non-tech with probability 0.7
 - become unemployed with probability 0.15
- A person who is **unemployed** (U) will
 - find a tech job with probability 0.25
 - find a non-tech job with probability 0.5
 - remain unemployed with probability 0.25

$$A = \begin{matrix} & \begin{matrix} T & N & U \end{matrix} \\ \begin{matrix} T \\ N \\ U \end{matrix} & \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} \end{matrix}$$

- a) Complete the transition matrix *A* above right.
- b) What is the probability that a person who is working in tech will be unemployed in three years?
- c) In the long run, what percentage of the population will be unemployed?
- d) If the region has 50 thousand workers, how many of them will be working in the tech industry in the long run?
- e) After 4 years there are 17.06, 25.48 and 7.46 thousand workers respectively in the three states. How many thousands of workers were in each state after 1 year?